UniVersHOI: Learning When to Act, Wait, and Act Together for Cooperative Human-Object Interaction

Anonymous CVPR submission

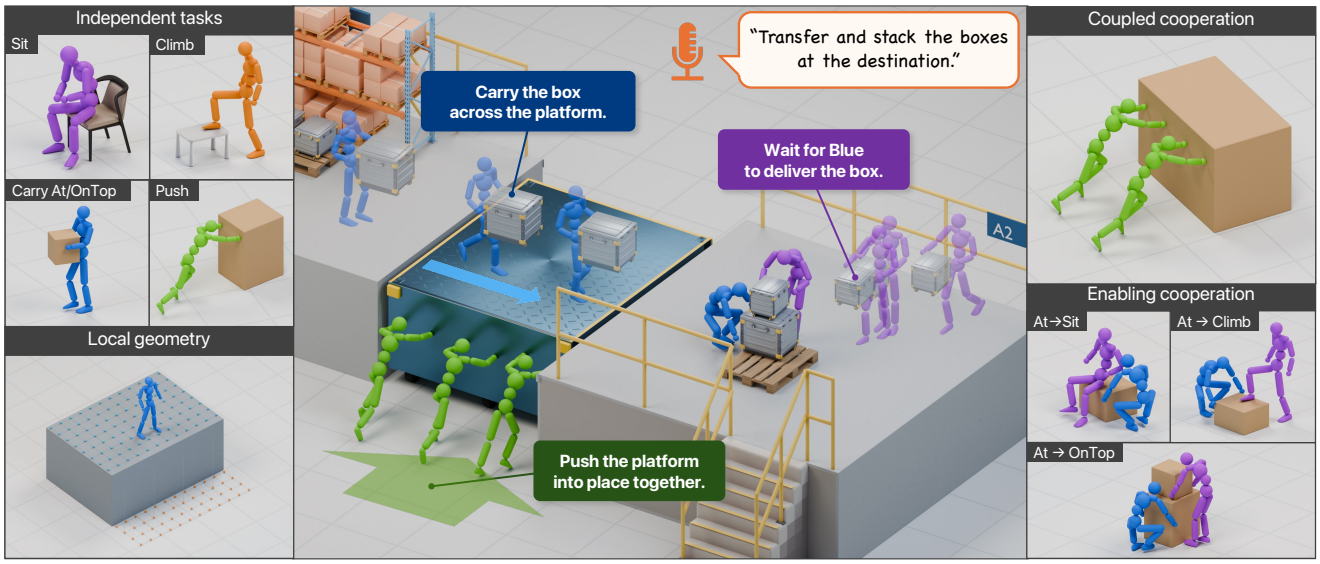

Figure 1. We present **UniVersHOI**, a single physics-based policy that unifies diverse interaction skills and inter-agent dependencies to perform independent, enabling, and coupled interactions across varying scene configurations without manually specified low-level coordination phases. With high-level task planning and local geometry observations, UniVersHOI executes complex cooperative scenarios.

Abstract

Consider the complex cooperative scenario in Fig. 1: several humanoids jointly push a platform into place to form a bridge, others carry boxes across it, and one waits for a box to arrive before stacking its own on top. Each agent performs a familiar skill, but whether it should act, wait, or act together depends on how its task relates to its teammates' tasks. Yet prior physics-based methods have largely developed multi-skill control and cooperative manipulation separately, and prescribe execution phases for dependent tasks. We present UniVersHOI, a single physics-based policy that learns when to act, when to wait, and when to act together across independent, enabling, and coupled interactions. Our key idea is to represent task assignments and inter-agent dependencies as relations that condition attention. A Scalable Scene-Centric Transformer (SCT) uses task-conditioned attention to jointly encode varying numbers of agents, objects, and goals. A Dependency-Aware Coordination Transformer (DCT) conditions inter-agent attention on dependencies between agents' tasks. Trained without cooperative motion demonstrations, the policy prepares for dependent tasks and sustains joint manipulation as the scene evolves, rather than following prescribed execution phases. It substantially outperforms hand-designed coordination rules and a cooperatively trained baseline with the same inputs and rewards, and generalizes from four-agent cooperative training to 16-agent scenes without retraining. With high-level LLM planning and local geometry adaptation, UniVersHOI executes complex cooperative scenarios.

1. Introduction

When people work together, what one person can do often depends on what the others do. In Fig. 1, a group of simulated humanoids pushes a platform into place, others

carry boxes across it, and one waits for a box to be delivered before stacking its own on top. Each agent performs a familiar interaction skill, but the scene succeeds only if each agent relates to the others in the right way: some must act in lockstep (*coupled*), some must wait for and build on a teammate's result (*enabling*), and the rest can proceed on their own (*independent*). Cooperation therefore hinges on a decision that no individual skill makes: when to act, when to wait, and when to act together.

Physics-based character control has advanced in both multi-skill control and cooperative manipulation, but largely along separate lines. Multi-skill methods give one humanoid a broad repertoire, such as sitting, climbing, and carrying, within a single policy [3, 24, 41]. Cooperative methods let several humanoids carry or push a shared object [2, 4, 19, 38, 42, 45], scaling to larger teams [19] and to sequential transport [9]. Most of them center on joint manipulation, *i.e.*, coupled interactions, while existing work on sequential cooperation uses a hand-designed state machine to prescribe execution phases and transitions, rather than allowing the policy to directly coordinate agents as the scene evolves. Bringing these capabilities together in a single policy that learns to coordinate agents under diverse inter-agent dependencies thus remains largely unexplored.

We aim for interaction versatility: handling diverse interaction skills and inter-agent dependencies together. To extend this versatility across scenes with varying numbers and configurations of agents and objects, we also need scene scalability. Achieving both within a single policy is hard for two reasons. First, each agent must identify the entities relevant to its skill across varying scene configurations. Second, and more fundamentally, the policy must be versatile in how agents relate, not only in what each one does. How an agent should treat a teammate depends on the relation between their tasks, not just on either agent's skill: a climber must hold back while a teammate places the box it will climb, yet be ready the moment the box lands; agents pushing one platform must move in lockstep; and an agent on an independent task need not coordinate at all. Common workarounds fall short: per-agent multi-skill policies ignore these relations, so a climber may start before its box arrives. Even with hand-designed coordination procedures, handling changing situations requires additional rules, such as letting a waiting agent move out of a teammate's path.

We present UniVersHOI, a single physics-based policy that performs diverse interaction skills and handles independent, enabling, and coupled interactions across varying scene configurations. Our key idea is to represent both task assignments and inter-agent dependencies as relations that condition attention. A Scalable Scene-Centric Transformer (SCT) encodes the scene as a variable set of agent, object, and goal tokens and conditions attention on task relations, in the manner of relation-aware transformers [27, 43], so that each agent focuses on the entities relevant to its skill. A Dependency-Aware Coordination Transformer (DCT) applies the same principle to inter-agent attention using dependency relations. Given task assignments and dependency relations, the policy determines when to act, when to wait, and when to act together based on the evolving scene state, without explicitly prescribed coordination phases.

We train the policy with reinforcement learning [26] and an adversarial motion prior [25], without cooperative motion demonstrations. We evaluate the same trained policy across varying scene sizes, including mixed scenes of independent, enabling, and coupled interactions. For complex cooperative scenarios, we combine UniVersHOI with high-level LLM planning and local geometry adaptation, as shown in Fig. 1.

Our main contributions are summarized as follows:

- We formulate cooperative multi-humanoid interaction through diverse interaction skills and three interaction types (independent, enabling, and coupled), and present UniVersHOI, a single physics-based policy that learns when to act, wait, and act together without manually specified coordination phases.
- We introduce SCT to encode variable-sized scenes and condition attention on task relations for scalable multi-skill control, and DCT to condition inter-agent attention on dependency relations for dependency-aware coordination.
- We show that UniVersHOI learns to prepare for dependent tasks and sustain joint manipulation, outperforming dependency-unaware, rule-based, and cooperatively fine-tuned baselines (the latter with the same inputs and rewards) across all three interaction types, generalizes to 16-agent scenes without retraining, and supports complex cooperative scenarios.

2. Related Work

2.1. Versatile Human-Object Interaction

Recent work on physics-based human-object interaction (HOI) has explored unified policies for diverse interaction tasks [29, 30, 32]. To achieve physically plausible interactions, PhysHOI [33] uses contact-graph rewards, while InterMimic [40] distills specialized imitation teachers into a unified controller. SkillMimic [34] learns reusable interaction skills, and SkillMimic-V2 [44] improves their robustness and transitions through demonstration augmentation. TokenHSI [24] takes a different approach, bringing heterogeneous skills together through composable task tokens. To make interaction control more flexible, other methods incorporate scene context [7], language-guided contact planning [37], and VLM-guided goal and reward design [3]. Generative interaction priors [41] and unified scene interaction and manipulation [12, 16] further extend these capabilities.While these methods offer versatile control for individual humanoids, extending such versatility to scenes

with varying numbers of agents, objects, and goals poses a distinct challenge. Our method addresses this through a scalable scene representation with task-relation-conditioned attention, supporting reliable individual skill execution across multi-humanoid scenes.

2.2. Multi-Agent Interaction

Prior work on multi-agent interaction spans kinematic motion generation and physics-based character control. Kinematic approaches [5, 10, 17, 23] synthesize coordinated motions from language or observed partner motions. ReGenNet [39] captures asymmetric action-reaction dynamics, while InterControl [35] guides generation with pairwise joint-distance constraints. Physics-based studies [11, 31, 36, 45] investigate multi-character interactions in simulation. InterAgent [14] models fine-grained joint-to-joint spatial dependencies through interaction graphs, while AssistMimic [28] learns partner-aware policies for contact-rich assistive motion tracking. These approaches capture inter-agent coupling through body motion and physical contact. Our work instead focuses on dependency relations between tasks assigned to different humanoids, distinguishing independent, enabling, and coupled interactions. By explicitly modeling these relations, our policy coordinates diverse individual skills across varying multi-agent scenes.

2.3. Cooperative Human-Object Interaction

Cooperative motion modeling has advanced through interaction datasets and benchmarks [20], synchronized motion generation [8], contact-aware motion refinement [13], stability-driven motion generation with physical refinement [38], and interaction-preserving retargeting [18]. In physics-based control, Zhang et al. [45] reproduce cooperative carrying through interaction-graph-based imitation, while CooHOI [4] transfers individual interaction skills to cooperative control through shared object dynamics. TeamHOI [19] further supports varying team sizes through a decentralized policy with teammate tokens. Recent methods explore skill transfer and reuse in SynAgent [42] and From Solo to Ensemble [2], while Commander-Executor [9] combines high-level planning with physics-based control for cooperative object manipulation. Prior studies largely focus on shared-object manipulation, with some sequential approaches relying on hand-designed phases and transitions [9]. Our unified policy instead models dependency relations across assigned tasks to coordinate independent, enabling, and coupled interactions without prescribed coordination phases.

3. Method

3.1. Overview

Our goal is to develop a unified physics-based policy for multi-agent HOI that supports both interaction versatility and scene scalability. We denote the shared scene observation at time t by S_t, which contains information about objects, goals, and N humanoid agents. Each agent is assigned a task defined by an interaction skill and its associated entities. Task relations $\mathcal{R}_{\text{task}}$ connect agents with task-relevant objects and goals, while dependency relations $\mathcal{R}_{\text{dep}}$ specify dependencies among the assigned tasks. Given these inputs, a single policy generates physical actions for all N agents:

$$(a_{1,t}, \ldots, a_{N,t}) = \pi_\theta(S_t, \mathcal{R}_{\text{task}}, \mathcal{R}_{\text{dep}}), \tag{1}$$

where each action $a_{i,t}$ specifies joint targets for the corresponding agent. The relations specify who does what and which tasks depend on which, not when to act. As illustrated in Fig. 2, the Scalable Scene-Centric Transformer (SCT) jointly encodes scene entities through entity self-attention, conditioned on task relations, to produce interaction-aware agent representations. The Dependency-Aware Coordination Transformer (DCT) then applies agent self-attention to these representations, conditioned on dependency relations. A shared action head maps the resulting representations to physical actions. We use proximal policy optimization (PPO) [26] with an adversarial motion prior (AMP) [25] for SCT pretraining and cooperative policy training.

3.2. Scalable Scene-Centric Transformer

Shared Scene Representation. Given S_t, we consider three entity types: agents, objects, and goals. Each entity u is represented by a feature vector $\mathbf{x}_u$ and a pose T_u. The pose T_u specifies the entity's position and orientation in a shared world coordinate frame. The contents of $\mathbf{x}_u$ depend on the entity type: agent features encode articulated states, including body poses and velocities; object features capture motion and bounding-box geometry; and goal features are constant, with target locations specified by their poses. Each feature vector is then mapped to a common embedding space by a type-specific MLP and combined with a learnable type embedding to form an entity token. Both the encoders and type embeddings are shared across entities of the same type.

The resulting tokens are jointly processed by a shared transformer through entity self-attention. To incorporate the relative positions and orientations of entities into attention, we adopt Geometric Transform Attention (GTA) [22]. Using entity poses T_u, GTA transforms queries, keys, and values from each entity's local frame into a shared world frame for attention, then maps the outputs back to their local frames. Detailed transformations are provided in the supplementary material. SCT thus produces representations for all agents in a single forward pass without separately encoding the scene for each agent. With shared entity encoders and variable-length self-attention, SCT accommodates varying numbers of agents, objects, and goals without architectural changes.

Task Attention Bias. We consider five interaction skills: SIT, CLIMB, AT (carrying and placing an object at a tar-

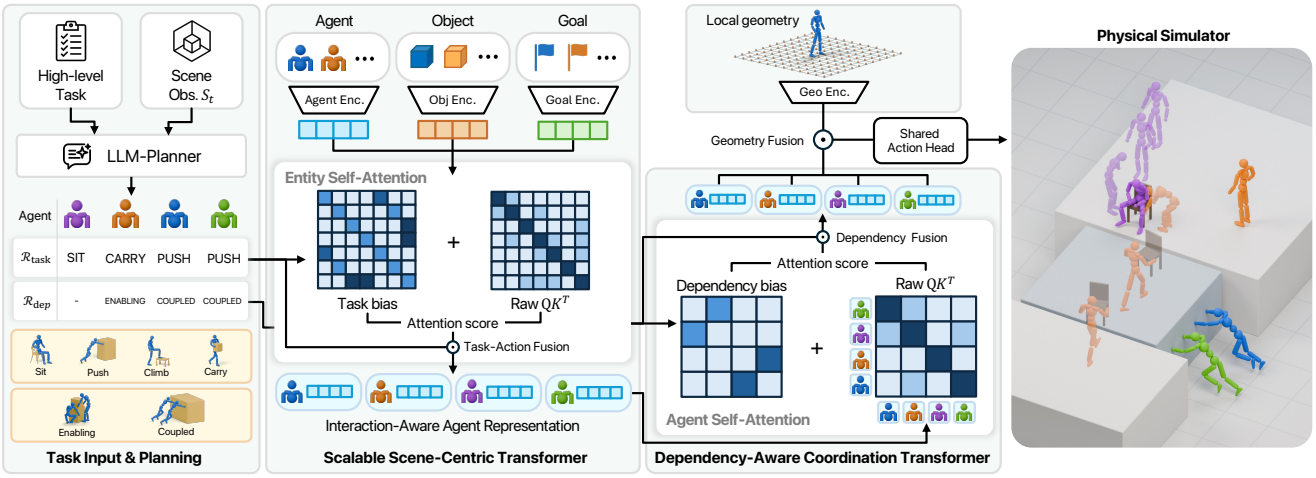

Figure 2. **Overview of UniVersHOI.** Given scene observations and task and dependency relations, the Scalable Scene-Centric Transformer (SCT) jointly encodes agents, objects, and goals using task attention biases and task fusion. The Dependency-Aware Coordination Transformer (DCT) then incorporates inter-agent dependencies through dependency attention biases and dependency fusion to generate coordinated actions through a shared action head. High-level task planning and local geometry adaptation extend UniVersHOI to complex cooperative scenarios.

get location), ONTOP (carrying and placing an object onto another object), and PUSH (pushing an object to a target location). Based on $\mathcal{R}_{\text{task}}$, we construct directed connections among tokens associated with each agent's assigned task. For example, SIT connects an agent token to the object the agent sits on, while AT connects the agent to the manipulated object and goal, and the object to the goal.

For each task connection associated with skill s, we compute a scalar attention bias:

$$b_s = \mathbf{p}^\top \mathbf{e}_s. \tag{2}$$

Here, $\mathbf{e}_s$ is a learned skill embedding, and $\mathbf{p}$ is a projection vector learned separately for each transformer layer and attention head. These biases form $\mathbf{B}_{\text{task}}$, with zero bias for unconnected token pairs. The entity self-attention weights are

$$\mathbf{A} = \text{softmax}\left(\frac{\widetilde{\mathbf{Q}}\widetilde{\mathbf{K}}^\top}{\sqrt{d}} + \mathbf{B}_{\text{task}}\right), \tag{3}$$

where $\widetilde{\mathbf{Q}}$ and $\widetilde{\mathbf{K}}$ are GTA-transformed queries and keys, and d is the head dimension.

Task Fusion. While task attention biases guide information exchange among scene entities, task fusion directly conditions each agent's representation on its assigned skill. At each transformer layer, the post-attention feature $\mathbf{z}_i$ of agent i is updated through a residual MLP:

$$\mathbf{z}_i \leftarrow \mathbf{z}_i + f_{\text{task}}\left([\mathbf{z}_i; \mathbf{e}_{s_i}]\right). \tag{4}$$

Here, s_i denotes the skill assigned to agent i, and $[\cdot;\cdot]$ denotes concatenation. The skill embeddings $\mathbf{e}_s$ are shared with the task attention bias, and f_{task} is shared across agents and transformer layers. The final agent-token outputs form the interaction-aware representations $\mathbf{H} = [\mathbf{h}_1, \ldots, \mathbf{h}_N]^\top$, which serve as input to DCT.

We first train the scene-centric policy on the five interaction skills in two-agent scenes. Across all skills, task rewards comprise progress, state, and success terms, encouraging movement toward the objective, attainment of the desired physical state, and task completion, respectively.

3.3. Dependency-Aware Coordination Transformer

Dependency Attention Bias. Building on SCT's task-conditioned agent representations $\mathbf{H}$, DCT applies agent self-attention to model inter-agent dependencies specified by $\mathcal{R}_{\text{dep}}$. We consider two cooperative dependency types, ENABLING and COUPLED. ENABLING connects a follower's query to its predecessor's key, while COUPLED establishes bidirectional connections among agents jointly manipulating an object. Agents performing independent interactions have no assigned inter-agent dependencies.

For each dependency connection of type r, we compute an attention bias from its learned dependency embedding $\mathbf{e}_r$:

$$b_r = \mathbf{p}^\top \mathbf{e}_r. \tag{5}$$

Here, $\mathbf{p}$ is a DCT-specific projection learned for each layer and attention head, and $\mathbf{B}_{\text{dep}}$ assigns zero bias to agent pairs without dependencies. The agent self-attention weights are

$$\mathbf{A} = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}} + \mathbf{B}_{\text{dep}}\right), \tag{6}$$

where $\mathbf{Q}$ and $\mathbf{K}$ are queries and keys computed from agent representations. Both task and dependency biases modulate attention without masking unconnected pairs.

Dependency Fusion. While dependency attention biases guide inter-agent information exchange, dependency fusion directly conditions agent attention contexts on dependency types. At each transformer layer, the attention context of each ENABLING follower or COUPLED participant is updated through a residual MLP:

$$\mathbf{m}_i \leftarrow \mathbf{m}_i + f_{\text{dep}}\left([\mathbf{m}_i; \mathbf{e}_r]\right). \tag{7}$$

Here, $\mathbf{m}_i$ denotes the attention-weighted value feature before the output projection, and the dependency embeddings are shared with the dependency attention bias. The MLP f_{dep} is shared across agents, dependency types, and transformer layers. Agents without applicable dependencies bypass dependency fusion. The DCT outputs form coordinated agent representations $\mathbf{C} = [\mathbf{c}_1, \ldots, \mathbf{c}_N]^\top$, which are mapped to physical actions through a shared action head.

Enabling and Coupled Cooperation. For enabling cooperation, a predecessor places an object at a target location (AT), allowing a follower to perform SIT, CLIMB, or ONTOP using the placed object. To support the follower's preparation, we use a virtual object token that retains the physical object's features and orientation but replaces its position with the predecessor's target position. The follower's task connections refer to this token, while physical interactions occur with the actual object. For coupled cooperation, multiple agents jointly push a shared object. Each participant's PUSH task is connected to a separate object token representing the same physical rigid body. SCT accommodates both object-token configurations within the same architecture, while DCT models the corresponding inter-agent dependencies.

Cooperative Policy Training. We separately train cooperative policies for enabling and coupled interactions in four-agent scenes. The pretrained SCT actor encoder is frozen, while DCT and the shared action head are optimized for each policy. We then distill these policies into a unified policy for independent, enabling, and coupled interactions.

For enabling cooperation, the follower's progress reward remains active, while its state and success rewards are enabled only when the predecessor's AT objective is satisfied. A movement penalty discourages the follower from moving within a waiting region before AT completion. For coupled cooperation, each agent is rewarded for approaching the shared object, while rewards for moving it toward the goal become active only when all participants are positioned to push. These rewards specify what each dependency requires rather than a schedule of execution phases. Detailed reward formulations and training settings for both stages are provided in the supplementary material.

3.4. Cooperative Scenario Execution

For cooperative scenario execution, we extend UniVersHOI with high-level task planning and local geometry adaptation. An LLM planner converts a high-level instruction and scene information into task relations $\mathcal{R}_{\text{task}}$ and dependency relations $\mathcal{R}_{\text{dep}}$. While the planner determines the task structure, the learned policy decides when each agent acts, without manually specified low-level coordination phases. Task and dependency relations remain fixed during execution.

To account for geometric constraints, we sample surface heights on an agent-centered, heading-aligned grid for each agent and encode the resulting height map using an MLP. The geometry feature is concatenated with the corresponding DCT representation and passed to a geometry-conditioned action head shared across agents. We freeze SCT and DCT and train the geometry encoder and action head for scenario execution. Detailed planner and geometry adaptation settings are provided in the supplementary material.

4. Experiments

We first evaluate unified control of diverse interaction skills and inter-agent dependencies (interaction versatility) and generalization beyond the training scale (scene scalability) (Sec. 4.2), then analyze when the policy acts, waits, and acts together (Sec. 4.3). We examine the contributions of task and dependency fusion (Sec. 4.4) and demonstrate integrated execution in complex cooperative scenarios (Sec. 4.5).

4.1. Experimental Setup

Implementation Details. We implement our framework by extending the simulation environment provided by TokenHSI [24]. We use reference motions from AMASS [21], SAMP [6], OMOMO [15], and BONES-SEED [1] for AMP [25] and reference state initialization (RSI), without direct motion tracking or cooperative motion demonstrations.

Baselines. We adapt TokenHSI [24] to our five interaction skills and shared multi-agent setting, denoted as TokenHSI*. For multi-agent execution, TokenHSI* controls each agent independently using agent-specific observations from the shared scene. We also compare against the pretrained SCT, which jointly processes the shared scene without explicit dependency-aware coordination.

To evaluate different coordination strategies, we consider rule-based and learned approaches. SCT + FSM (finite-state machine) applies hand-designed waiting and release rules without additional policy training. For enabling interactions, the follower is frozen until the predecessor completes AT; for coupled interactions, each agent is frozen upon contact with the shared object, and all participants are released once everyone establishes contact. SCT + Coop. FT replaces DCT with an MLP that aggregates features from partners and other agents, and is trained using the same input information and cooperative rewards as Ours. All methods use the same task assignments, with equivalent virtual-object information for enabling interactions and shared-object information for coupled interactions.

Table 1. **Interaction versatility and scene scalability.** Independent (Ind.) and Mixed SR (%, mean ± std. over three random seeds).

	2 agents		4 agents		8 agents		16 agents	
Method	Ind.	Mixed	Ind.	Mixed	Ind.	Mixed	Ind.	Mixed
TokenHSI*	85.84±1.11	56.29±0.98	83.51±0.82	51.48±1.51	80.96±0.62	50.10±0.69	82.06±0.59	51.02±0.70
SCT	90.36±0.75	64.05±1.38	88.51±0.34	62.16±2.04	88.48±0.13	57.75±0.39	81.12±0.26	46.66±0.89
SCT + FSM	90.36±0.75	71.01±1.33	88.51±0.34	67.69±0.29	88.48±0.13	64.39±1.41	81.12±0.26	52.13±0.44
SCT + Coop. FT	93.85 ± 0.39	79.27 ± 1.15	93.86 ± 0.60	75.27 ± 1.75	92.90 ± 0.61	71.98 ± 0.84	84.94 ± 0.27	59.37 ± 0.86
Ours	**97.40** ± 0.65	**92.54** ± 0.95	**96.55** ± 0.45	**89.09** ± 0.97	**94.87** ± 0.27	**85.45** ± 1.06	**90.15** ± 0.28	**71.41** ± 0.27

Evaluation Metrics. We measure task completion using success rate (SR) with task-specific spatial criteria. For AT and PUSH, success requires the physical object center to be within 20 cm of its target in 3D space, consistent with the carrying-task distance threshold in TokenHSI [24]. For SIT, CLIMB, and ONTOP, success requires the pelvis (SIT/CLIMB) or payload center (ONTOP) to be within the supporting object's horizontal bounds and 20 cm of the target height. We evaluate 512 episodes per random seed across three seeds for each setting, with a maximum of 900 control steps (30 s) per episode.

We compute SR at the interaction-group level: independent interactions are evaluated per agent, enabling pairs succeed when the predecessor's AT and follower's task conditions are simultaneously satisfied, and coupled groups succeed when their shared object reaches the goal. For mixed scenes, we report the macro-average

$$\mathrm{SR}_{\mathrm{Mixed}} = \frac{\mathrm{SR}_I + \mathrm{SR}_E + \mathrm{SR}_C}{3}, \tag{8}$$

where SR_I, SR_E, and SR_C denote the success rates of independent, enabling, and coupled groups in mixed scenes, respectively, with equal weight given to each interaction type. Additional details on implementation, reference motions, training, baseline adaptations, and evaluation protocols are provided in the supplementary material.

4.2. Interaction Versatility and Scene Scalability

Tab. 1 evaluates Independent and Mixed SR across scenes with 2, 4, 8, and 16 agents, including configurations beyond our four-agent cooperative training scale.

TokenHSI* applies the same multi-skill policy separately to each agent, whereas SCT jointly processes all agents, objects, and goals within a shared scene. However, SCT's limited Mixed SR shows that unified scene modeling alone is insufficient for effective cooperation. SCT + FSM improves Mixed SR over SCT through hand-designed coordination rules, while SCT + Coop. FT achieves further gains through cooperative training using the same input information and rewards as Ours. Ours achieves the highest Independent and Mixed SR across all evaluated scene sizes. At 16 agents, it achieves 90.15% Independent SR and 71.41% Mixed SR, outperforming the second-best method, SCT + Coop. FT, by 12.04 percentage points in Mixed SR. This gap reflects the benefit of conditioning inter-agent attention on dependencies in DCT, rather than of additional information or rewards. Fig. 3 shows independent, enabling, and coupled interactions executed concurrently in a 16-agent scene. Trained for cooperation with four agents, UniVersHOI generalizes to 8- and 16-agent scenes without additional fine-tuning, generating actions for all agents in a single forward pass.

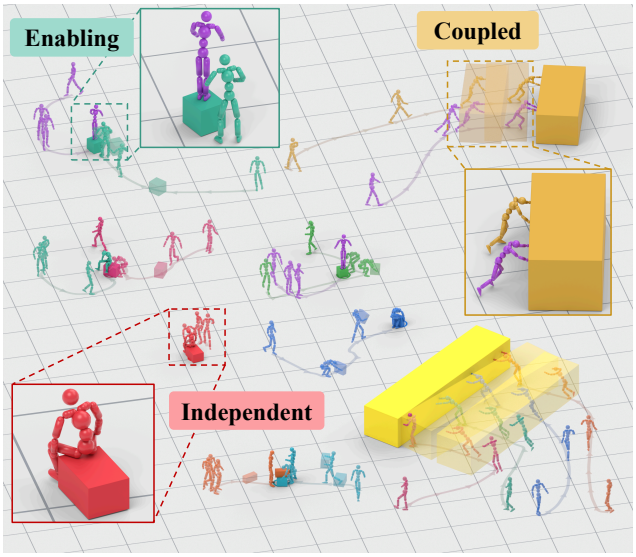

Figure 3. **Mixed-scene execution with 16 agents.** UniVersHOI executes independent, enabling, and coupled interactions in a shared scene. Overlaid poses and trajectories show agent motion over time.

Table 2. **Enabling cooperation.** Group SR (%) for AT→SIT/CLIMB/ONTOP. Time (Succ./All) reports mean completion time over successful trials and all trials, respectively, with failures counted as 30 s.

Method	AT→SIT↑	AT→CLIMB↑	AT→ONTOP↑	Avg.↑	Time (Succ./All) (s)↓
TokenHSI*	14.19	5.73	11.26	10.39	**9.90**/27.91
SCT	19.01	35.29	21.09	25.13	12.29/25.60
SCT + FSM	45.77	43.88	28.91	39.52	14.79/23.99
SCT + Coop. FT	63.09	54.36	52.86	56.77	12.20/19.90
Ours	**86.85**	**85.29**	**81.97**	**84.70**	10.64/**13.60**

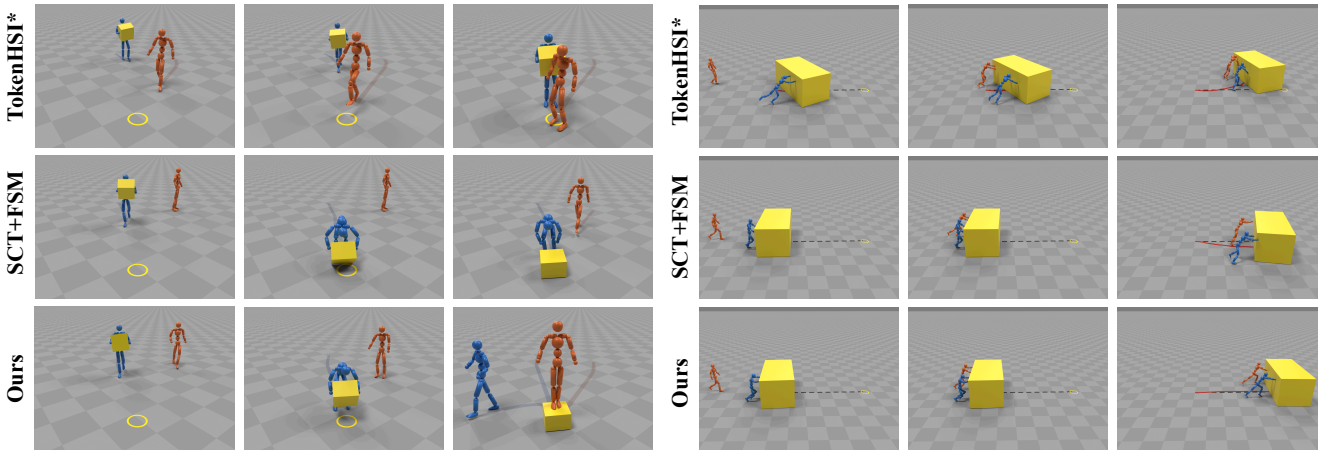

Figure 4. **Qualitative comparison of enabling and coupled coordination.** Rows show TokenHSI*, SCT + FSM, and Ours (top to bottom). Columns show three frames in temporal order. (a) Blue: predecessor; orange: follower. (b) Blue contacts the object before orange. Black dashed lines show straight paths to goals. Red solid lines show actual object trajectories. Ours approaches in advance without obstructing placement (a) and sustains joint pushing (b).

Table 3. **Coupled cooperation.** Group SR (%) by number of agents and cooperative time ratio t_{coop} (%, pooled across trials), adapted from TeamHOI [19].

Method	2-agent SR↑	3-agent SR↑	4-agent SR↑	Avg.↑	t_{coop} ↑
TokenHSI*	73.05	62.63	53.52	63.06	46.61
SCT	80.08	74.02	69.47	74.52	53.31
SCT + FSM	83.85	72.14	69.40	75.13	62.87
SCT + Coop. FT	76.04	73.05	72.53	73.87	80.43
Ours	**95.18**	**92.19**	**87.57**	**91.64**	**84.05**

4.3. When to Act, Wait, and Act Together

Enabling cooperation. Enabling cooperation requires the follower to prepare for its dependent task without interfering with the predecessor. Tab. 2 shows that Ours achieves the highest average Group SR of 84.70%, exceeding the second-best method by 27.93 percentage points, along with the lowest mean completion time across all trials (13.60 s). Although TokenHSI* records a slightly shorter completion time on successful trials, its success rate is much lower. SCT + FSM improves Group SR over SCT through rule-based waiting, but remains well below Ours and takes longer on both successful and all trials. Fig. 4(a) shows these behavioral differences. The TokenHSI* follower occupies the placement location before the object arrives and obstructs the predecessor, while SCT + FSM delays the follower's approach until AT completion. In contrast, Ours approaches in advance, waits without obstructing placement, and executes its dependent task after AT completion. These results highlight the importance of coordinating preparation without interfering with the predecessor, beyond rule-based waiting.

Coupled cooperation. Coupled cooperation requires agents not only to synchronize the start of pushing but also to coordinate their actions throughout manipulation. Since Group SR measures goal attainment without requiring all assigned agents to participate, we additionally report the cooperative time ratio t_{coop}, measuring the fraction of pushing time with all participants simultaneously in contact. Tab. 3 shows that Ours achieves the highest average Group SR of 91.64%, exceeding the second-best method by 16.51 percentage points, along with the highest t_{coop} of 84.05%. Although SCT + FSM synchronizes the start of pushing and SCT + Coop. FT substantially increases simultaneous contact through partner-aware cooperative training, neither achieves a substantial improvement in Group SR over SCT. Fig. 4(b) shows these behavioral differences. TokenHSI* begins pushing before its teammates arrive, while SCT + FSM synchronizes the start of pushing but the object subsequently deviates from its goal. In contrast, Ours waits for its teammates and sustains joint pushing until the object reaches its goal. These results highlight the benefit of dependency-aware coordination for successful joint manipulation while maintaining simultaneous contact. Notably, these coordinated behaviors are learned without cooperative motion demonstrations or manually specified low-level coordination phases.

4.4. Ablation Study

We investigate whether feature fusion provides additional benefits beyond attention biases. Tab. 4 evaluates task fusion in SCT and dependency fusion in DCT using Independent and Mixed SR, respectively. Removing task fusion substantially reduces Independent SR, with the performance drop increasing from 23.63 percentage points in 2-agent scenes to 73.07 points in 16-agent scenes, demonstrating its im-

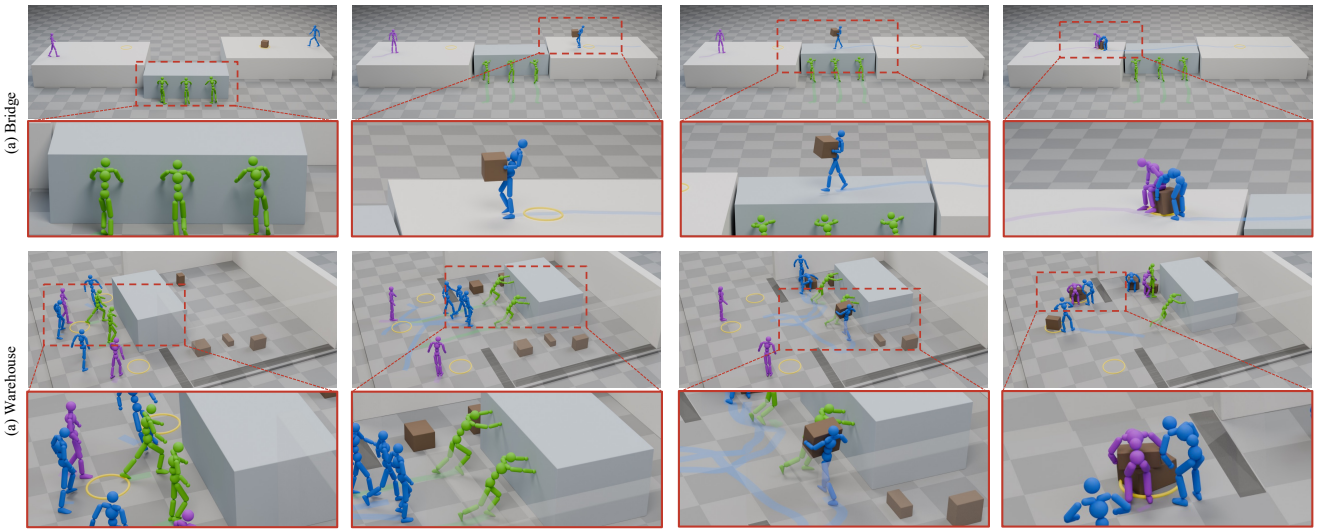

Figure 5. **Cooperative scenario execution.** (a) Bridge: Joint PUSH establishes a traversable bridge, followed by object delivery and CLIMB. (b) Room: Joint PUSH clears a passage, followed by object delivery and SIT.

Table 4. **Feature fusion ablation.** Task fusion in SCT and dependency fusion in DCT are evaluated using Independent and Mixed SR (%), respectively.

Evaluation	Method	2-agent SR↑	4-agent SR↑	8-agent SR↑	16-agent SR↑
Independent	w/o Task Fusion	66.73	59.91	33.67	8.05
	SCT	**90.36**	**88.51**	**88.48**	**81.12**
Mixed	w/o Dep. Fusion	85.36	81.51	78.05	59.18
	Ours	**92.54**	**89.09**	**85.45**	**71.41**

portance for maintaining multi-skill performance as scene size increases. Removing dependency fusion reduces Mixed SR across all scene sizes, with the largest drop of 12.23 percentage points in 16-agent scenes, confirming its additional contribution to cooperative coordination. Even without dependency fusion, dependency attention biases alone outperform SCT + Coop. FT in 2- to 8-agent scenes (Tab. 1).

4.5. Cooperative Scenario Execution

Fig. 5 demonstrates two complex cooperative scenarios integrating coupled and enabling interactions. In Bridge, multiple agents jointly push a platform into place to form a traversable bridge, allowing others to deliver objects and perform subsequent dependent tasks. In Room, joint pushing clears a blocked passage, enabling other agents to deliver objects and complete their assigned interactions. For these scenarios, an LLM planner provides only task and dependency assignments, while the geometry-adapted policy uses local geometry observations for physical execution and decides when each agent acts, waits, or acts together, without manually specified low-level coordination phases. Additional analysis is provided in the supplementary material.

5. Conclusion

We presented UniVersHOI, a unified physics-based policy that learns when to act, when to wait, and when to act together from the dependencies between agents' tasks, across diverse interaction skills and varying scene configurations. By combining task-aware scene modeling through SCT with dependency-aware coordination through DCT, our policy generates coordinated actions for all agents without manually prescribed execution phases. Experiments demonstrate substantial improvements over independent, rule-based, and cooperatively fine-tuned baselines, with generalization to scenes of up to 16 agents without retraining. UniVersHOI further learns distinct coordination behaviors for enabling and coupled interactions, supporting timely preparation and sustained joint manipulation. We demonstrate the integration of these capabilities in complex cooperative scenarios with high-level planning and local geometry adaptation. Since task and dependency relations currently remain fixed during execution, updating them online, *e.g.*, through replanning, is a natural next step.

LLM Usage Statement

We used ChatGPT(OpenAI) to assist with language refinement, including improvements to grammar, clarity, and coherence. We also used it for coding assistance, such as code review, debugging, and refinement. All AI-assisted suggestions were reviewed by the authors, who take full responsibility for the final work.

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